

Supplemental Material for “Emergent $\mathcal{PT}$ Symmetry and Exceptional Points in a Driven Dirac Impurity”

Vinayak M. Kulkarni¹

¹*Theoretical Sciences Unit, Jawaharlal Nehru Centre for Advanced Scientific Research, Bangalore 560064, India*
(Dated: July 17, 2026)

This Supplemental Material provides the technical derivations and validation checks that support the self-contained main article. It includes the auxiliary-mode influence functional, Keldysh mean-field equations, physical-FDR construction, shifted-resonance condition, frozen SW/BA-kernel algebra, interaction-regularized EP-distance diagnostic, HWHM validity tests, negative signed-weight diagnostics, and supplementary figures.

NOTATION AND PARAMETER CONVENTIONS

- $t_{\text{av}}, \tau, T_{\text{b}}$: Wigner center time $(t + t')/2$, relative time $t - t'$, and reservoir temperature.
- η, ζ_{σ} : Chiral bath index $\eta = \pm$ and spin sign $\zeta_{\uparrow} = +1, \zeta_{\downarrow} = -1$.
- λ, j_{α} : SOC strength and eigenvalues of the dimensionless exchange matrix $\rho_b J^{\text{bio}}$; j_{scr} is the largest positive real part.
- $\alpha_{\text{b}}, k_{\text{max}}, D_{\text{uv}}$: Quadratic bath-dispersion coefficient, radial momentum cutoff, and ultraviolet energy cut-off. The two cutoffs are set numerically to $\pi/4$ in the dimensionless plots but are not the same physical quantity.
- $W_k, W, \Lambda_{\eta}^{\text{PV}}$: Static bare impurity-bath hybridization and principal-value bath shift.
- $\beta_0, \tilde{\beta}$: Signed frozen drive-control amplitude and nonnegative SBMF projection $\tilde{\beta} = |\beta_0| |b_c|$. The bare analytic and full projected EP controls are $\beta_{0,\text{bare}}^{\text{EP}} = 0.50$ and $\beta_{0,\text{proj}}^{\text{EP}} = 0.4975$, respectively.
- s, s_{eff} : Bare and interaction-regularized impurity splittings; ζ_{σ} is used for the spin sign.
- $\gamma, \Gamma_{\text{PT}}, \bar{\Gamma}$: Bare coherent impurity flip, passive relative decay imbalance, and common bath width. The symbols $\bar{\Gamma}_{\eta}$ and $\Gamma_{\sigma}^{\text{net}}$ denote, respectively, the chiral frozen-kernel width and the physical net channel width.
- ω, Ω, ω_0 : Fourier frequency, drive frequency, and a selected spectral pole or resonance frequency; none of these denotes temperature.

- $\mathcal{E}_n, E_{\text{EP}}, \Lambda(\omega)$: Non-Hermitian eigenvalues, EP eigenvalue, and real bath self-energy.
- $\mathcal{W}_{\text{FSR}}^+, \mathcal{W}_{\text{CP}}^+$: Positive spectral weights in shifted-resonance and central-peak windows.
- $\mathcal{P}, \mathbf{a}, \mathbf{b}$: Full-kernel Schur polynomial and the two local contact denominators; these are distinct from the bath coefficient α_{b} .
- $F(\lambda), \mathcal{F}_{\text{dist}}, \mathcal{F}_{\text{MF}}$: SOC-overlap form factor, distribution matrix, and mean-field free-energy functional.
- $D_{\text{occ}}, \mathcal{D}_{\text{EP}}$: Doubly occupied impurity amplitude and EP-boundary discriminant.

Appendix A: Impurity-block EP condition and condition-number scaling

This appendix records the impurity-block algebra used in the main text. The Hadamard rotation itself is shown in Eqs. (38)–(39); here we keep only the EP and condition-number scaling that underlies the diagnostic interpretation.

The frozen impurity block is $h_d = \begin{bmatrix} \tilde{\epsilon}_{\xi} + \gamma & i\beta_0 \\ i\beta_0 & \tilde{\epsilon}_{\xi} - \gamma \end{bmatrix}$, with eigenvalues $E_{\pm} = \tilde{\epsilon}_{\xi} \pm s$ and splitting $s = \sqrt{\gamma^2 - \beta_0^2}$. The regimes are $\mathcal{PT}$ -unbroken ($\gamma > \beta_0$), exceptional ($\gamma = \beta_0$), and $\mathcal{PT}$ -broken ($\beta_0 > \gamma$). Right eigenvectors may be chosen as $|r_{\pm}\rangle \propto [\beta_0, i(\gamma \mp s)]^T$, so

$$\det R = 2i\beta_0 s, \quad R = (|r_{+}\rangle, |r_{-}\rangle). \quad (\text{A1})$$

Thus, near the bare EP,

$$\kappa_{\text{imp}} = \|R\| \|R^{-1}\| \sim \frac{C}{|s|}, \quad (\text{A2})$$

with a nonsingular constant C . In the interacting frozen diagnostic, this divergence is cut off by the regularized splitting $s_{\text{eff}} > 0$ (App. K), giving a finite EP-distance diagnostic rather than a thermodynamic scale by itself.

Appendix B: Angular Harmonic Decomposition and 2D→1D Reduction

We show how the 2D Dirac bath with trigonal anisotropy reduces to an effective 1D problem after integrating out all angular-momentum harmonics $m \neq 0$. This Gaussian reduction is the microscopic origin of the 1D BA structure in Sec. IX.

1. Angular Fourier Decomposition

In 2D polar momentum coordinates (k, θ) , the bath fermion is decomposed into angular-momentum channels:

$$c_{k,\theta,\sigma} = \frac{1}{\sqrt{2\pi}} \sum_{m=-\infty}^{\infty} c_{k,m,\sigma} e^{im\theta}. \quad (\text{B1})$$

The impurity at position $\mathbf{r} = 0$ (the lattice site) is local, so only the isotropic ($m = 0$) channel hybridizes with it:

$$c_{\mathbf{r}=0,\sigma} = \sum_k \frac{1}{\sqrt{2\pi}} c_{k,0,\sigma}. \quad (\text{B2})$$

Higher harmonics $m \neq 0$ are decoupled from the impurity in the absence of anisotropy.

2. C_3 Selection Rule from $\cos 3\theta$

The trigonal anisotropy $\beta_3(k)k^3 \cos 3\theta$ is expanded as $\cos 3\theta = \frac{1}{2}(e^{3i\theta} + e^{-3i\theta})$. Substituting Eq. (B1) and performing the θ integral: $\int_0^{2\pi} \frac{d\theta}{2\pi} e^{i(m'-m)} e^{\pm 3i\theta} = \delta_{m',m \mp 3}$. The trigonal term therefore couples only channels differing by $\Delta m = \pm 3$:

$$V_{\text{trig}} \propto \sum_{k,\sigma,m} c_{k,m,\sigma}^\dagger c_{k,m-3,\sigma} + \text{h.c.} \quad (\text{B3})$$

The auxiliary fermions $O_{k,m,\sigma}$ represent precisely the $m = \pm 3, \pm 6, \dots$ channels activated by this selection rule.

3. Gaussian Integration and Passive Relative Kernel

Since $H_O + H_{\text{hyb}}$ is bilinear in $O_{k,m,\sigma}$, integrating out all $m \neq 0$ modes is a Gaussian path integral within the quadratic auxiliary sector. For fixed (k, θ, σ) the auxiliary action has exactly the form

$$\begin{aligned} S_O + S_{cO} = & \int_c dt dt' \bar{O}(t) A_O(t, t') O(t') \\ & + \int_c dt [\bar{c}(t) V_\sigma(t) O(t) + \bar{O}(t) V_\sigma^\dagger(t) c(t)] \end{aligned} \quad (\text{B4})$$

The Grassmann integral gives the exact quadratic influence action

$$S_{\text{infl}} = - \int_c dt dt' \bar{c}(t) V_\sigma(t) g_O(t, t') V_\sigma^\dagger(t') c(t'), \quad (\text{B5})$$

so the explicit retarded vertex contraction is

$$\Sigma_\sigma^{O,R}(t, t') = \sum_{m, m' \neq 0} V_{m\sigma}(t) g_{mm'}^{O,R}(t, t') V_{m'\sigma}^*(t'). \quad (\text{B6})$$

No anomalous auxiliary propagator is used. For a number-conserving Hermitian auxiliary Hamiltonian this contraction contains the phase difference fixed by the vertex and its Hermitian conjugate. Therefore the off-shell auxiliary sector produces a real principal-value correction in the low-energy window. This conclusion is also immediate after the unitary rotation $O_{k,m,\sigma} \rightarrow e^{-i\zeta_\sigma \phi(t)} O_{k,m,\sigma}$, which removes the drive phase from the vertices and shifts the auxiliary levels by $-\zeta_\sigma \dot{\phi}(t)$.

The auxiliary-induced correction is therefore a coherent spin-odd shift, not an anti-Hermitian relative gain-loss term:

$$\begin{aligned} \Sigma_\sigma^{\text{bath},R}(\omega, t_{\text{av}}) = & \Sigma_0^{\text{bath},R}(\omega) + \zeta_\sigma \Delta_{\text{aux}}(\omega, t_{\text{av}}) \\ & + O[(\Omega/\epsilon_O)^2], \quad \Delta_{\text{aux}} \in \mathbb{R}. \end{aligned} \quad (\text{B7})$$

The passive non-Hermitian component appears only after this shifted channel is transmitted through the retarded Dirac propagator. The point is clearest from the quadratic (d, c, O) inverse Green-function kernel,

$$\mathcal{G}^{-1} = \begin{pmatrix} g_d^{-1} & -W^\dagger & 0 \\ -W & g_c^{-1} & -V_\sigma \\ 0 & -V_\sigma^\dagger & g_O^{-1} \end{pmatrix}. \quad (\text{B8})$$

Here W is static and all explicit drive dependence resides in $V_\sigma(t)$. Integrating out O and then c gives the two exact Schur complements

$$G_{c,\sigma}^{-1} = g_c^{-1} - V_\sigma g_O V_\sigma^\dagger, \quad \Sigma_{d,\sigma} = W^\dagger G_{c,\sigma} W. \quad (\text{B9})$$

Equivalently,

$$\Sigma_{d,\sigma} = W^\dagger g_c W + W^\dagger g_c (V_\sigma g_O V_\sigma^\dagger) g_c W + \dots, \quad (\text{B10})$$

which explicitly displays the drive-dependent virtual path $d \xrightarrow{W} c \xrightarrow{V_\sigma(t)} O \xrightarrow{V_\sigma^\dagger(t')} c \xrightarrow{W^\dagger} d$. The order of the two Gaussian integrations is immaterial.

For $\Xi_\sigma^R = \Xi_0^R + \zeta_\sigma \delta \Xi^R$, the general linear correction is

$$\delta \Sigma_d^R = W^\dagger G_{c,0}^R \delta \Xi^R G_{c,0}^R W. \quad (\text{B11})$$

It is complex because $G_{c,0}^R$ is a retarded continuum propagator, even when the off-shell $\delta \Xi^R$ is real. In the rigid-shift limit $\delta \Xi^R \simeq \Delta_{\text{aux}}$, with weak frequency dependence of the spin-even auxiliary background,

$$\Sigma_{d,\sigma}^R(\omega, t_{\text{av}}) \simeq \Sigma_{d,0}^R(\omega - \zeta_\sigma \Delta_{\text{aux}}) \simeq \Sigma_{d,0}^R - \zeta_\sigma \Delta_{\text{aux}} \partial_\omega \Sigma_{d,0}^R. \quad (\text{B12})$$

Writing $\Sigma_{d,0}^R = \Lambda - i\bar{\Gamma}$, the real derivative gives $\Delta_{\text{coh}} = -\Delta_{\text{aux}} \partial_\omega \Lambda$ in the σ_z channel—it does *not* renormalize the spin-mixing splitting $\Delta_{\text{eff}\sigma_x}$ at this order—while the imaginary derivative gives

$$\Gamma_{\text{PT}} = \Delta_{\text{aux}} \partial_\omega \bar{\Gamma}(\omega_0). \quad (\text{B13})$$

Here ω_0 denotes the low-energy evaluation point of the frozen kernel; at the compensation energy where $\partial_\omega \Lambda$ changes sign (of order D_{uv}/e for the linear Dirac width with a sharp cutoff), the Kramers–Kronig detuning vanishes, $\Delta_{\text{coh}}(\omega_0) = 0$, while $\partial_\omega \bar{\Gamma}$ remains finite. The SOC split Dirac bath provides the needed helicity channels and the energy-dependent width

$$\bar{\Gamma}_\eta(\omega) = \pi \sum_k |W_{k\eta}|^2 \delta[\omega - \varepsilon_{c,\eta}(k)]. \quad (\text{B14})$$

After subtracting the common damping $-i\bar{\Gamma}I$, the remaining co-decaying kernel contains the passive- $\mathcal{PT}$ core $\Delta_{\text{eff}}\sigma_x + i\Gamma_{\text{PT}}\sigma_z$ of Eq. (31) together with the $\mathcal{PT}$ -odd Kramers–Kronig detuning $\Delta_{\text{coh}}\sigma_z$; exact passive $\mathcal{PT}$ symmetry holds at the compensated point $\Delta_{\text{coh}} = 0$, and a finite detuning acts as the causal floor of Eq. (33). This is the sense in which the strictly Hermitian driven model generates the relative non-Hermitian kernel used in the main text. If the bath width is flat, $\partial_\omega \bar{\Gamma} = 0$, or if the helicity channels are degenerate, the same auxiliary correction produces only a coherent spin splitting and no passive $\mathcal{PT}$ decay imbalance.

The reduction relies on the following conditions:

- (i) Impurity at a C_3 -invariant site ($\theta_{\text{imp}} = 0 \bmod 2\pi/3$), so the angular projection leaves a single radial bath channel.
- (ii) Off-shell auxiliary harmonics, $\epsilon_O^{(m)} \gg D_{uv}$, so the auxiliary contribution is a controlled principal-value shift. Residual on-shell weight gives ordinary bath damping rather than the relative $\mathcal{PT}$ channel.
- (iii) Slow drive on the auxiliary memory time, $\Omega \ll \epsilon_O^{(m)}$, so the correction is organized in powers of Ω/ϵ_O .
- (iv) A bath with energy-dependent width, such as the spin-orbit-split Dirac bath, so that $\partial_\omega \bar{\Gamma} \neq 0$.
- (v) H_O bilinear: no many-body approximation is made in the auxiliary-mode integration.

This completes the microscopic link asserted at the start of the appendix. The key point is that the Gaussian auxiliary projection gives a real spin-odd shift, and the retarded Dirac bath converts that shift into a passive relative gain–loss component in the co-decaying frame, together with its Kramers–Kronig companion detuning; the SOC off-diagonal bath dressing [Eq. (28)] supplies the $\mathcal{PT}$ -even coherent mixing channel.

a. Result: 2D bath $\rightarrow$ 1D effective bath. After the Gaussian integration, the impurity is coupled to only the $m = 0$ bath channel, labeled by radial momentum k alone. The 1D effective bath retains the Dirac density of states $\rho_{\text{bath}}(\omega) \sim |\omega|/\lambda^2$ because the 2D Jacobian $\sim k$ from the angular integration combines with the quadratic dispersion $\epsilon_k = \alpha_b k^2$ to give a linear-in- $|\omega|$ DOS.

The Gaussian reduction leaves a single radial ($m = 0$) Dirac bath carrying the linear DOS $\rho_{\text{bath}}(\omega) \sim |\omega|/\lambda^2$,

hybridizing with the impurity through the channel-preserving vertex of Eq. (B2). After the spin rotation of Sec. IV B this radial bath splits into the two chiral channels $c_{k\pm}$, each coupled diagonally to a single impurity component $\tilde{d}_\pm$. This one-dimensional, channel-diagonal effective bath is precisely the structural input consumed by the frozen biorthogonal Bethe-Ansatz construction of Sec. IX: the block-diagonal S -matrix [Eq. (63)] and the right/left sector separation follow directly from the diagonal hybridization produced here, so that no additional bath structure is introduced between the Gaussian reduction and the frozen BA diagnostic.

Appendix C: Keldysh Mean-Field Theory

The slave-boson representation $d_\sigma^\dagger = f_\sigma^\dagger b$ introduces spinon fermions f_σ (the Kondo quasiparticles) and a slave boson b (enforcing no double occupancy) with condensate $b_c = \langle b \rangle$ and Lagrange multiplier δ_c enforcing the constraint $r^2 + \sum_\sigma f_\sigma^\dagger f_\sigma = Q$ (where $r = |b_c|$). Here $Q = 1$ is the physical spin-1/2 constraint, while $Q = Nq_{\text{occ}}$ denotes the large- N or channel-rescaled normalization used in the corresponding numerical saddle. The substitution is defined in the physical spin- $\uparrow\downarrow$ basis; the self-consistent equations are derived by varying the Keldysh action with respect to b_c and δ_c in this basis, giving the stationarity condition and constraint in the $\uparrow\downarrow$ representation:

$$b_c \delta_c + 2i\beta_0 \sum_{k,\sigma} G_{f_\sigma, c_\sigma}^< = 0, \quad (\text{C1})$$

$$r^2 - i \sum_{k,\sigma} [G_{\text{ss}}^<(t, t)]_{f_\sigma, f_\sigma} = Q. \quad (\text{C2})$$

These are then rewritten in the rotated $\tilde{f}_\pm$ basis by applying the Hadamard rotation $\tilde{f}_\pm = (f_\uparrow \pm f_\downarrow)/\sqrt{2}$ (the spinon analog of $\tilde{d}_\pm$ in the main text). At the Keldysh saddle point $b_q = \delta_q = 0$ (where subscript q denotes the quantum/Keldysh component), varying the action with respect to the classical component of the boson gives the equivalent $\pm$ -basis form:

$$b_c \delta_c + 2i\beta_0 \sum_k [G_{\tilde{f}_+, c_+}^< + G_{\tilde{f}_-, c_-}^<] = 0, \quad (\text{C3})$$

$$r^2 - i \sum_{k,\pm} [G_{\text{ss}}^<(t, t)]_{\tilde{f}_\pm, \tilde{f}_\pm} = Q, \quad (\text{C4})$$

where $\tilde{f}_\pm = (f_\uparrow \pm f_\downarrow)/\sqrt{2}$ are the rotated spinons. The rewriting from Eqs. (C1)–(C2) to the $\pm$ basis follows from two algebraic facts: (i) the Hadamard rotation is unitary, so $\sum_\sigma G_{f_\sigma, c_\sigma}^< = G_{\tilde{f}_+, c_+}^< + G_{\tilde{f}_-, c_-}^<$ and $\sum_\sigma [G^<]_{f_\sigma, f_\sigma} = \sum_\pm [G^<]_{\tilde{f}_\pm, \tilde{f}_\pm}$; (ii) the diagonal hybridization structure of $\tilde{H}(k)$ [Eq. (40) of the main text] forces $G_{\tilde{f}_+, c_-}^< = G_{\tilde{f}_-, c_+}^< = 0$ within the diagonal-channel saddle point, so no cross-channel terms contribute. Equation (C3) is the stationarity condition for b_c ; Eq. (C4)

is the constraint. The lesser Green's functions are evaluated using $G^< = f(\omega)[G^R - G^A]$ for the bath-controlled bilinear reservoir coupling [1]. The causal constraint on the Lagrange multiplier is implemented by projecting $\delta_c \leftarrow \text{Re}(\delta_c)$ at each iteration, which is equivalent to enforcing the retarded (causal) boundary condition.

a. Numerical implementation. The numerics work directly with $H_{ss}(k)$ [Eq. (37)] in the mixed basis $(d_\uparrow, d_\downarrow, c_{k+}, c_{k-})$ —the impurity in the physical $\uparrow\downarrow$ spin basis and the bath in the chiral $\pm$ basis (the bath diagonalization from Sec. II is already done). The Hadamard rotation to $(\tilde{d}_+, \tilde{d}_-, c_{k+}, c_{k-})$ is *not* applied numerically; instead the saddle-point quantities are accumulated directly as $\mathcal{S}_c = \sum_k (G_{d_\uparrow, c_+}^< + G_{d_\downarrow, c_+}^<)$ and $\mathcal{C}_m = \sum_k (G_{d_\uparrow, c_-}^< + G_{d_\downarrow, c_-}^<)$ from the full 4×4 biorthogonal lesser Green's function, and the update equations for δ_c and r follow from $\delta_{c, \text{new}} = \text{Re}[(-2i\beta_0 \mathcal{S}_c + 2\beta_0 \mathcal{C}_m)/b_c]$ and $r_{\text{new}}^2 = \text{Re}(Q + i\mathcal{X}_d)$ with $\mathcal{X}_d = \sum_k (G_{d_\uparrow, d_\uparrow}^< + G_{d_\downarrow, d_\downarrow}^<)$. The numerical gauge is fixed by choosing $b_c = r > 0$. For the two-channel diagnostic implementation one sets the same rescaled Q used in Eqs. (C2) and Eq. (C4). These are the $\uparrow\downarrow$ -basis saddle-point equations [Eqs. (C1)–(C2)], and they are numerically equivalent to the $\pm$ -basis form by the unitary equivalence established above.

The numerical implementation uses the gauge-fixed amplitude $\tilde{\beta} = |\beta_0||b_c|$ and then constructs the projected channels as $\Gamma_{PT} = c_T \tilde{\beta}$, $\Delta_{\text{eff}} = \Delta_0 + c_\Delta \tilde{\beta}$, and $V_{\text{eff}} = c_V \tilde{\beta}$. The symmetric representative is $\Delta_0 = 0$ and $c_T = c_\Delta = c_V = 1$; generic channel-projected data may use $\Delta_0 \neq 0$ or unequal slopes to realize an isolated finite-drive impurity EP. This is a low-energy projection convention for the diagnostic Hamiltonian; it is not obtained by allowing the phase of b_c to enter H_{ss} .

The cross-channel contribution $\mathcal{C}_m$ vanishes identically in the $\mathcal{PT}$ -unbroken phase by the block-diagonal structure of the S -matrix [Eq. (63)]: the $-V$ sign in the $(d_\downarrow, c_-)$ hybridization entry of H_{ss} ensures $G_{d_\uparrow, c_-}^< = -G_{d_\downarrow, c_-}^<$ to high precision, so $\mathcal{C}_m \approx 0$ and the dominant contribution comes from $\mathcal{S}_c$ alone.

Numerical solution gives the self-consistent amplitude $|b_c|$ after fixing the slave-boson gauge. The phase of b_c is a $U(1)$ gauge choice; only the amplitude $|b_c|$ is physically determined and used in H_{ss} . Gauge checks are performed by repeating the saddle-point iteration from real, negative-real, and complex initial seeds for $b_c^{(0)}$ and comparing the gauge-invariant outputs $|b_c|$, $\text{Re}(\delta_c)$, $\tilde{\beta}$, and $\text{Re}(\tilde{\epsilon}_\xi)$.

b. Note on SOC parameter for Figs. 1 and 2. Figures 1 and 2 use $\lambda = k_{\text{max}} = \pi/4$, which places the system at the edge of the SOC Lifshitz transition where $F(\lambda) = 0$ [Eq. (K4)]. In this limit the quartic (App. K) gives $s_{\text{eff}} = 0$ at the bare EP, so the EP in panel (a) is the *unregularised* bare EP; the interaction-regularised EP ($s_{\text{eff}} > 0$, $F > 0$) is demonstrated in Fig. 5 where $\lambda = 0.50 < k_{\text{max}}$.

Appendix D: Fluctuation-Dissipation Relation: Two Constructions

Two constructions of the lesser Green's function arise naturally in a non-Hermitian driven system, and it is important to distinguish their physical content.

a. Bath-based (physical). $G_{\text{bath}}^<(\omega) = f(\omega)[G^R(\omega) - G^A(\omega)]$ follows directly from the Keldysh identity for a system bilinearly coupled to a thermal reservoir [1]. It satisfies the bath-controlled FDR for this bilinear reservoir setup, independent of whether H is Hermitian. In the diagonal-hybridization basis of $\tilde{H}$, the two channels $\tilde{d}_\pm$ are independently coupled to the $c_{k\pm}$ baths, and each satisfies the FDR: $[G_{\text{bath}}^<]_{\pm\pm} = f(\omega)[G^R - G^A]_{\pm\pm}$.

b. Eigenmode-based (comparison). $G_{\text{eig}}^< = iR \text{diag}(f(\text{Re } \mathcal{E}_n)) L^\dagger$ assigns thermal Fermi occupations to the real parts of the (complex) non-Hermitian eigenvalues $\mathcal{E}_n$. This does not correspond to the bath-imposed occupations; the deviation $G_{\text{eig}}^< - G_{\text{bath}}^<$ grows with $|\text{Im}(\mathcal{E}_n)|$ and diverges near the EP. In the uniform convention of Fig. 4 ($\epsilon_\xi = -U/2 = -1.0$, $U = 2.0$, $T_b = 0.1$), the gain eigenvalue acquires an imaginary part $|\text{Im}(\mathcal{E}_n)|$ that grows with the gauge-fixed scale $\tilde{\beta} = |\beta_0|r$ and becomes comparable to the coherent splitting as $\beta_0 \rightarrow \beta_{0, \text{proj}}^{\text{EP}} = 0.4975$. The eigenmode deviation is therefore of order $|\text{Im}(\mathcal{E}_n)|/T_b$ before multiplication by the thermal factor $f(\text{Re } \mathcal{E}_n) \sim 1$, and it diverges as the EP is approached and the left/right eigenvectors become collinear. This growth is the origin of the large excursions of the eigenmode ratio in Fig. 4(c); it is a signature of eigenvector nonorthogonality in the non-Hermitian representation, not of a physical distribution function. The bath-controlled ratio in panel (a), by contrast, follows $f_{\text{FD}}(\omega)$ to within the exported residual and is the only object used for occupations and SBMF self-consistency.

c. Correct DOS. For non-Hermitian systems, the physical spectral function is $A_{\text{imp}}(\omega) = -\frac{1}{2\pi} \text{Im Tr}[G^R - G^A]_{\text{imp}} \geq 0$. Using only $-\text{Im Tr } G^R/\pi$ introduces spurious negative signed contributions of the relative co-decaying kernel; the antisymmetrized combination $G^R - G^A$ cancels them in the $\mathcal{PT}$ -unbroken phase [2].

d. Mode-resolved diagnostic matrix. One can formally define an effective distribution matrix $\mathcal{F}_{\text{dist}}(\omega) = G^<(\omega)[G^R(\omega) - G^A(\omega)]^{-1}$. For the bath-based construction $\mathcal{F}_{\text{dist}} = f(\omega)I$ in the reservoir-coupled basis. For the eigenmode construction, $\mathcal{F}_{\text{dist}}$ is generally matrix-valued and frequency dependent. This is why the eigenmode comparison in Fig. 4 is treated only as a nonnormality diagnostic, while all physical occupations and SBMF equations use $G_{\text{bath}}^<$.

Appendix E: Shifted Friedel-Type Resonance Condition

This appendix gives the conservative form of the shifted resonance condition used in Sec. VD. The purpose is not to claim a new exact sum rule, but to identify the reference position of the low-energy peak within the SBMF Green's function.

1. Resonance Position in the Lorentzian Regime

The spinon retarded Green's function at the SBMF saddle point may be written as

$$G_{f,\sigma}^R(\omega) = \frac{1}{\omega - \tilde{\epsilon}_\xi + i\Gamma_\sigma^{\text{net}}}, \quad \Gamma_\sigma^{\text{net}} = \Gamma_\sigma - x_\sigma, \quad (\text{E1})$$

where $x_\sigma = \zeta_\sigma \Gamma_{\text{PT}} = \pm \Gamma_{\text{PT}}$ is the spin-selective imaginary shift from the coarse-grained drive and $\Gamma_\sigma^{\text{net}} > 0$ in the stable low-energy regime. The corresponding physical impurity spectral function is

$$A_{\text{imp}}(\omega) = \frac{1}{\pi} \sum_\sigma \frac{\Gamma_\sigma^{\text{net}}}{(\omega - \text{Re}(\tilde{\epsilon}_\xi)^2 + (\Gamma_\sigma^{\text{net}})^2)}. \quad (\text{E2})$$

When the line shape is approximately Lorentzian, both spin channels are centred at the same reference frequency

$$\omega_{\text{res}} \simeq \text{Re}(\tilde{\epsilon}_\xi). \quad (\text{E3})$$

This is the shifted Friedel-type spectral resonance condition used in the figures. In the strongly non-Lorentzian regime, the observed maximum can deviate from this reference; those deviations are treated as line-shape effects rather than as a breakdown of a sum rule.

2. Origin of the Shift

In the Hermitian particle-hole-symmetric SBMF problem, $\epsilon_\xi = -U/2$ and the spin occupations are equal, so the saddle point pins $\text{Re}(\tilde{\epsilon}_\xi)$ near zero. In the driven problem the coarse-grained self-energy gives unequal effective widths $\Gamma_\uparrow^{\text{net}} \neq \Gamma_\downarrow^{\text{net}}$ and a real projected shift δ_Σ , which together move the saddle value through

$$\text{Re}(\tilde{\epsilon}_\xi) = \epsilon_\xi + \text{Re}(\delta_c) + \delta_\Sigma. \quad (\text{E4})$$

The shift is therefore a gauge-invariant consequence of the spin-selective self-energy and of the SBMF saddle point. It is not attributed to the phase of b_c .

a. Numerical reference. For the frozen channel-projected kernel used in the figures, $\text{Re}(\tilde{\epsilon}_\xi)$ is the renormalized impurity level generated by the same plotted data pipeline. In the local-moment reference with $\epsilon_\xi = -U/2 = -1.0$, the frozen closure gives $\text{Re}(\tilde{\epsilon}_\xi) \simeq -1.0$. If a separate finite-temperature self-consistent SBMF closure is used, Eq. (E4) should instead be evaluated with that closure and the figure should be regenerated from the same saddle.

Appendix F: Slave-Boson Mean-Field Kondo Scales

This appendix summarizes the slave-boson mean-field (SBMF) scale estimate in the rotated spinon basis. The emphasis is on the spin-selective passive relative decay shifts, the channel-resolved resonance width, and the way an EP-active Jordan residue can modify the screening kernel without implying a universal exact Kondo temperature.

1. Effective Spinon Hamiltonian

Using the slave-boson representation $d_\sigma^\dagger = f_\sigma^\dagger b$ with gauge-invariant condensate amplitude $r = |b_c|$, the quadratic effective Hamiltonian for the spinon fermions f_σ is:

$$H_{\text{eff}} = \sum_\sigma \tilde{\epsilon}_\sigma f_\sigma^\dagger f_\sigma + \sum_{k\sigma} [r V_{k\sigma} c_{k\sigma}^\dagger f_\sigma + \text{h.c.}], \quad (\text{F1})$$

where $V_{k\sigma}$ is the hybridization and $\tilde{\epsilon}_\sigma = \tilde{\epsilon}_\xi + ix_\sigma$ is the complex spinon level. The chiral spinon modes $\tilde{f}_\pm = (f_\uparrow \pm f_\downarrow)/\sqrt{2}$ are the same rotated basis used in App. C. The spin-dependent projected imaginary shift $x_\sigma \equiv \zeta_\sigma \Gamma_{\text{PT}} = \pm \Gamma_{\text{PT}}$ (with $\zeta_\uparrow = +1$, $\zeta_\downarrow = -1$) encodes the drive-induced passive relative gain/loss; the real part $\epsilon_{r,\sigma} = \text{Re}(\tilde{\epsilon}_\sigma) = \tilde{\epsilon}_\xi$ is the renormalized level common to both spins. The renormalized hybridization width is:

$$\Gamma_\sigma(\omega) = \pi r^2 \sum_k |V_{k\sigma}|^2 \delta(\omega - \epsilon_{k\sigma}) \equiv r^2 \Gamma_\sigma^{(0)}(\omega), \quad (\text{F2})$$

and the spinon retarded Green's function is:

$$G_{f,\sigma}^R(\omega) = \frac{1}{\omega - \tilde{\epsilon}_\sigma + i\Gamma_\sigma}. \quad (\text{F3})$$

2. Saddle-Point Equations at $T_b = 0$

The constraint $r^2 + \sum_\sigma \langle f_\sigma^\dagger f_\sigma \rangle = Q$ and the stationarity condition $\partial \mathcal{F}_{\text{MF}} / \partial \delta_c = 0$ give at $T_b = 0$:

$$Q - r^2 = \sum_\sigma \int_{-D_{\text{uv}}}^0 \frac{d\omega}{\pi} \frac{\Gamma_\sigma}{(\omega - \epsilon_{r,\sigma})^2 + \Gamma_\sigma^2}, \quad (\text{F4})$$

$$\delta_c = \sum_\sigma \int_{-D_{\text{uv}}}^0 \frac{d\omega}{\pi} \frac{(\omega - \epsilon_{r,\sigma}) \Gamma_\sigma}{(\omega - \epsilon_{r,\sigma})^2 + \Gamma_\sigma^2}. \quad (\text{F5})$$

These are the standard SBMF equations evaluated with the complex spinon level $\tilde{\epsilon}_\sigma = \epsilon_{r,\sigma} + ix_\sigma$.

3. Kondo-scale estimate near an EP

Within slave-boson mean-field theory the low-energy scale is set by the renormalized impurity width. With $r = |b_c|$ and $V_{\text{eff}} = rV$,

$$\tilde{\Gamma}_\sigma(0) = r^2 \Gamma_\sigma^{(0)}(0), \quad \Gamma_\sigma^{(0)}(0) = \pi V^2 \rho_{b,\sigma}(0). \quad (\text{F6})$$

Matching to the usual poor-man's-scaling form gives the channel estimate

$$T_{K,\sigma} \simeq D_{\text{uv}} \exp \left[-\frac{\pi |\tilde{\epsilon}_\sigma|}{2\Gamma_\sigma^{(0)}(0)} \right], \quad |\tilde{\epsilon}_\sigma| = \sqrt{\epsilon_{r,\sigma}^2 + x_\sigma^2}, \quad (\text{F7})$$

where $x_\sigma = \zeta_\sigma \Gamma_{\text{PT}}$ is the projected spin-selective imaginary shift. The sign difference $x_\uparrow = -x_\downarrow$ is essential for the $\mathcal{PT}$ structure, but it does not by itself split T_K in Eq. (F7), because only x_σ^2 enters. Distinct channel scales arise when the SOC-split bath gives $\Gamma_\uparrow^{(0)}(0) \neq \Gamma_\downarrow^{(0)}(0)$, or when the real detunings $\epsilon_{r,\sigma}$ differ.

The EP-active resolvent should be tied to the screening exchange channel rather than to the condition number alone. In the EP-active subspace,

$$G_{\text{EP}}(z) = \frac{A_{\text{EP}}}{z - E_{\text{EP}}} + \frac{B_{\text{EP}}}{(z - E_{\text{EP}})^2} + G_{\text{reg}}(z), \quad (\text{F8})$$

where $B_{\text{EP}} \neq 0$ is the Jordan double-pole residue. Away from the exact EP, this singularity is cut off by

$$E_{\text{IR}} = \max(|s_{\text{eff}}|, \Gamma_{\text{EP}}, T_{\text{b}}). \quad (\text{F9})$$

Here Γ_{EP} denotes the residual retarded decay/broadening of the EP-active block, so E_{IR} is an energy cutoff rather than a new eigenvalue. The projection of this residue onto the biorthogonal exchange matrix renormalizes only those channels that have positive screening eigenvalues. The corresponding scale is therefore

$$T_K^{\text{est}} = D_{\text{uv}} \exp \left[-\frac{1}{\chi_K j_{\text{scr}}} \right], \quad (\text{F10})$$

$$j_{\text{scr}} = \max \{ \text{Re } j_\alpha : j_\alpha \in \text{spec}(\rho_b J^{\text{bio}}), \text{Re } j_\alpha > 0 \}. \quad (\text{F11})$$

Here j_α are the eigenvalues of $\rho_b J^{\text{bio}}$ and $\chi_K = 1$ in the plotted dimensionless convention. The condition number $\kappa_{\text{imp}} \sim C/|s_{\text{eff}}|$ remains a useful diagnostic of nonnormality and of the EP cutoff, but it is not inserted directly as a multiplicative physical coupling. If the EP-active sector is a positive antiferromagnetic screening channel, the effective exponent in Eq. (F10) is reduced. If the EP instead comes from a negative scalar hybridization product $V^R V^L < 0$, it defines a frozen non-Hermitian EP/crossover scale but not, by itself, an enhanced thermodynamic Kondo temperature.

Appendix G: Contact Algebra and S -Matrix

This appendix derives the block-diagonal S -matrix [Eq. (63)] from the two-body Bethe wavefunction of the $\tilde{H}(k)$ model. We first construct the wavefunction explicitly, then project the Schrödinger equation onto each impurity sector to obtain the contact equations, and finally extract the S -matrix by eliminating the impurity amplitudes.

1. Impurity Hilbert Space and Scattering Geometry

The frozen impurity block h_d in $\tilde{H}(k)$ has four states: empty $|0\rangle$, singly occupied $|\tilde{d}_+\rangle$ and $|\tilde{d}_-\rangle$, and doubly occupied $|D_{\text{occ}}\rangle = \tilde{d}_+^\dagger \tilde{d}_-^\dagger |0\rangle$. We consider two-particle scattering: one bath electron (momentum k , chiral channel $\eta \in \{+, -\}$) impinging on the impurity.

Scattering geometry. In the right sector, bath + electrons are right-movers incident on the impurity site. The S -matrix element $S_{\eta\eta}^{RR}$ relates the outgoing amplitude $\Phi_\eta^{\text{out}}(k)$ to the incoming amplitude $\Phi_\eta^{\text{in}}(k)$ in the same channel: $\Phi_\eta^{\text{out}}(k) = S_{\eta\eta}^{RR}(k) \Phi_\eta^{\text{in}}(k)$.

2. Two-Particle Bethe Wavefunction

For an incoming bath electron in channel $\eta = +$ with momentum k , the two-particle Bethe wavefunction is:

$$\begin{aligned} |\Psi_+^R(k)\rangle &= \Phi_+^{\text{in}}(k) c_{k+}^\dagger |0\rangle \\ &+ e_{++}(k) \tilde{d}_+^\dagger c_{k+}^\dagger |0\rangle + e_{-+}(k) \tilde{d}_-^\dagger c_{k+}^\dagger |0\rangle \\ &+ D_{\text{occ}}(k) \tilde{d}_+^\dagger \tilde{d}_-^\dagger |0\rangle + \Phi_+^{\text{out}}(k) c_{k+}^\dagger |0\rangle. \end{aligned}$$

3. Contact Equations from $\tilde{H}|\Psi_+^R\rangle = E|\Psi_+^R\rangle$

Projecting the Schrödinger equation onto each sector, keeping the bath hybridizations $V_\pm$ distinct from the internal relative channel β_0 : Here E is the total two-particle energy. For compactness, define $\mathbf{a} = \epsilon_{d,+} - E$ and $\mathbf{b} = \epsilon_{d,-} - E$.

Sector $|\tilde{d}_+, k_+\rangle$:

$$\mathbf{a} e_{++} + i\beta_0 e_{-+} + V_+(\Phi_+ + D_{\text{occ}}) = 0. \quad (\text{G1})$$

Sector $|\tilde{d}_-, k_+\rangle$:

$$\mathbf{b} e_{-+} + i\beta_0 e_{++} = 0. \quad (\text{G2})$$

Incoming c_{k-} sectors:

$$\mathbf{b} e_{--} + i\beta_0 e_{+-} + V_-(\Phi_- + D_{\text{occ}}) = 0, \quad (\text{G3})$$

$$\mathbf{a} e_{+-} + i\beta_0 e_{--} = 0. \quad (\text{G4})$$

Doubly-occupied sector:

$$(2\tilde{\epsilon}_\xi + U - E) D_{\text{occ}} + \sum_k (V_+ e_{++} + V_- e_{--}) = 0. \quad (\text{G5})$$

4. S -Matrix from Renormalized Denominators

From Eq. (G2): $e_{-+} = -(i\beta_0/\mathbf{b}) e_{++}$. Substituting into Eq. (G1): $\tilde{\mathbf{a}} e_{++} = -V_+(\Phi_+ + D_{\text{occ}})$ with $\tilde{\mathbf{a}} = \mathbf{a} + \beta_0^2/\mathbf{b}$. Similarly, from Eq. (G4): $e_{+-} = -(i\beta_0/\mathbf{a}) e_{--}$; substituting into Eq. (G3): $\tilde{\mathbf{b}} e_{--} = -V_-(\Phi_- + D_{\text{occ}})$

with $\tilde{\mathbf{b}} = \mathbf{b} + \beta_0^2/\mathbf{a}$. The doubly-occupied amplitude D_{occ} is eliminated using Eq. (G5). Since e_{++} depends only on Φ_+^{in} and D_{occ} , and e_{--} only on Φ_-^{in} and D_{occ} , the outgoing amplitudes $\Phi_\eta^{\text{out}} = \Phi_\eta^{\text{in}} + e_{\eta\eta}$ satisfy $S_{+-} = S_{-+} = 0$ identically.

Appendix H: Biorthogonal Structure and Eigenvectors

The frozen impurity block is $h_d = \begin{bmatrix} \tilde{\epsilon}_\xi + \gamma & i\beta_0 \\ i\beta_0 & \tilde{\epsilon}_\xi - \gamma \end{bmatrix}$ with splitting $s = \sqrt{\gamma^2 - \beta_0^2}$ and eigenvalues $E_\pm = \tilde{\epsilon}_\xi \pm s$.

a. Right eigenvectors.

$$|r_+\rangle \propto \begin{pmatrix} \beta_0 \\ i(\gamma - s) \end{pmatrix}, \quad |r_-\rangle \propto \begin{pmatrix} \beta_0 \\ i(\gamma + s) \end{pmatrix}. \quad (\text{H1})$$

b. Left eigenvectors. Because h_d is complex symmetric, the left eigenvectors may be taken as the transposes of the right eigenvectors:

$$\langle l_+ | \propto (\beta_0 \ i(\gamma - s)), \quad \langle l_- | \propto (\beta_0 \ i(\gamma + s)). \quad (\text{H2})$$

Direct verification gives $\langle l_\pm | r_\mp \rangle = 0$ and

$$\langle l_+ | r_+ \rangle = 2s(\gamma - s), \quad \langle l_- | r_- \rangle = -2s(\gamma + s). \quad (\text{H3})$$

Thus the biorthogonal normalization diverges as $1/s$ at the EP:

$$\mathcal{M}_\pm \mathcal{N}_\pm = \frac{1}{\langle l_\pm | r_\pm \rangle} \sim \frac{1}{s} \xrightarrow{s \rightarrow 0} \infty. \quad (\text{H4})$$

consistent with $\kappa_{\text{imp}} \sim C/|s|$ [Eq. (A2)]. At the EP ($s \rightarrow 0$): $|r_+\rangle \propto |r_-\rangle$, eigenvectors coalesce, and the Jordan block emerges with $h_d|r_+\rangle = E_{\text{EP}}|r_+\rangle$, $(h_d - E_{\text{EP}})|j\rangle = |r_+\rangle$.

1. Explicit Contact Projector and R-Matrix

The normalized rank-one projector in the $\{\tilde{d}_+, \tilde{d}_-\}$ contact space is

$$P_+ = \frac{|r_+\rangle\langle l_+|}{\langle l_+ | r_+ \rangle} = \frac{1}{2s(\gamma - s)} \begin{pmatrix} \beta_0^2 & i\beta_0(\gamma - s) \\ i\beta_0(\gamma - s) & -(\gamma - s)^2 \end{pmatrix}, \quad (\text{H5})$$

with $P_+^2 = P_+$. The full Hamiltonian $\tilde{H}(k)$ contains additional diagonal bath dispersions $\varepsilon_{c,\pm}(k)$ that do not enter the contact projector; they contribute only one-body propagation phases in the Bethe equations, leaving the YBE structure intact.

Appendix I: Yang–Baxter Equation

In each sector $X \in \{R, L\}$, the frozen two-body scattering matrix is a scalar Breit–Wigner amplitude in the

channel-separated contact problem. Equivalently one may write the contact part as

$$S^X(u) = \mathbb{I} + \chi^X(u)P^X, \quad (\text{I1})$$

where $P^X = |r^X\rangle\langle l^X|/\langle l^X | r^X \rangle$ is the normalized biorthogonal contact projector and $(P^X)^2 = P^X$. This projector notation is only a representation of the same scalar channel amplitude; no additional Baxterization functional equation is assumed.

Since the S -matrix in each frozen channel is a c-number, the three-particle Yang–Baxter equation is satisfied identically by commutativity of scalar multiplication:

$$S_{\eta\eta}^X(k_1 - k_3)S_{\eta\eta}^X(k_2 - k_3)S_{\eta\eta}^X(k_1 - k_2) = S_{\eta\eta}^X(k_1 - k_2)S_{\eta\eta}^X(k_2 - k_3)S_{\eta\eta}^X(k_1 - k_3). \quad (\text{I2})$$

Thus factorization in the frozen effective model follows from channel separation and scalar commutativity. We use this biorthogonal Bethe–Ansatz construction as an analytic diagnostic of the frozen channel-diagonal effective model rather than as a claim about the full self-consistent nonequilibrium impurity problem.

Appendix J: Biorthogonal Bethe Ansatz Equations

The charge rapidities $\{k_j^R\}$ satisfy the right-sector quantization condition [Eq. (69)]. The auxiliary rapidities $\{\mu_a^{R(\eta)}\}$ satisfy Eq. (70). Left-sector equations follow with $R \rightarrow L$ and $\mu^R \rightarrow \mu^L$. By pseudo-Hermitian symmetry $\mu^L = \overline{\mu^R}$, so only the right-sector equations are independent. For $U = 0$, the bare channel splitting $2|s|$ vanishes at $\beta_0 = \gamma$. For $U > 0$ and $F > 0$, the interaction-corrected $2s_{\text{eff}}$ remains finite at that control point, giving avoided coalescence rather than an interaction-shifted exact EP. Jordan-block structures in integrable models have been studied in Refs. [3, 4].

Appendix K: Interaction-Induced EP Shift

The interaction U shifts the bare EP via the self-consistent equation for the interaction-regularized splitting s_{eff} : Here and below, $F \equiv F(\lambda)$ denotes the SOC-overlap form factor.

$$s_{\text{eff}}^4 - s_0^2 s_{\text{eff}}^2 - U^2 \beta_0^2 F = 0, \quad s_0 = \sqrt{\gamma^2 - \beta_0^2}, \quad (\text{K1})$$

with positive root $s_{\text{eff}}^2 = \frac{1}{2}(s_0^2 + \sqrt{s_0^4 + 4U^2 \beta_0^2 F})$. At the bare EP ($s_0 = 0$): $s_{\text{eff}} = (U^2 \beta_0^2 F)^{1/4} = (U \beta_0 \sqrt{F})^{1/2}$ (non-analytic in U at the EP). At fixed β_0 , the positive root $s_{\text{eff}} > 0$ therefore lifts the bare degeneracy whenever $U > 0$ and $F > 0$. We do not reinterpret this algebraic splitting as a second definition of the coherent flip amplitude.

a. Origin of the EP-boundary diagnostic $\mathcal{D}_{\text{EP}} = \gamma^2 - C_F U^2 F$. Equation (K1) governs s_{eff} at *fixed* β_0 as U varies, answering: “does U regularise the EP?” (answer: yes, $s_{\text{eff}} > 0$). The EP-boundary diagnostic $\mathcal{D}_{\text{EP}}$ in Fig. 6 asks a complementary question: “for what (U, λ) is the system already in the PT-broken phase before β_0 is tuned to the bare EP?” This is determined by the self-consistent correction to the *flip amplitude* γ itself. At the bare EP ($\beta_0 = \gamma$, $s_0 = 0$) the off-diagonal impurity Green’s function develops a Jordan-block double pole,

$$G_{d_+, d_-}(\omega) = \frac{-i\gamma}{(\omega - \tilde{\epsilon}_\xi)^2}, \quad (\text{K2})$$

the hallmark of eigenvector coalescence [cf. App. H, Eq. (H4)]. The bath-mediated Fock correction to γ is $\delta\gamma = UF \text{Re} \sum_k G_{d_+, d_-}(\omega_k)$ [Eq. (K2)]; since the bath sum over the double pole is negative (occupied states below $\tilde{\epsilon}_\xi$ contribute with opposite sign), $\gamma_F = \gamma + \delta\gamma < \gamma$. Writing the dimensionless bath double-pole integral as a positive coefficient C_F , the leading local boundary diagnostic is

$$\mathcal{D}_{\text{EP}}(U, \lambda) = \gamma^2 - C_F U^2 F(\lambda) + \mathcal{O}(U^3). \quad (\text{K3})$$

The plotted convention is $C_F = 1$; this fixes the normalization of the analytic diagnostic and is not asserted to be a universal coefficient. The two diagnostics address distinct approximations. Equation (K1) describes interaction regularization at fixed β_0 , whereas Eq. (K3) estimates a local displacement of the bare EP boundary through the Fock-renormalized flip γ_F . They are not combined as a single self-consistent equation in the present treatment.

The bath correlation factor $F = \langle f_+(k) f_-(k') \rangle$ measures the inter-channel Fermi surface overlap. For the quadratic Dirac bath $\varepsilon_{c,\pm}(k) = \alpha_b k^2 \pm \lambda k$:

$$F(\lambda) = \max\left(1 - \frac{\lambda^2}{k_{\text{max}}^2}, 0\right). \quad (\text{K4})$$

This vanishes at $\lambda = k_{\text{max}}$, producing the sharp vertical wall in the analytic (U, λ) EP-boundary diagnostic [Fig. 6(b)] beyond which this particular interaction correction to the impurity-sector EP is switched off.

SUPPLEMENTARY FIGURE NOTES

The following supplementary figures provide compact diagnostics that support the main text. Unless stated otherwise, they use the same convention as the main figures, $\epsilon_\xi = -U/2 = -1.0$ with $U = 2.0$.

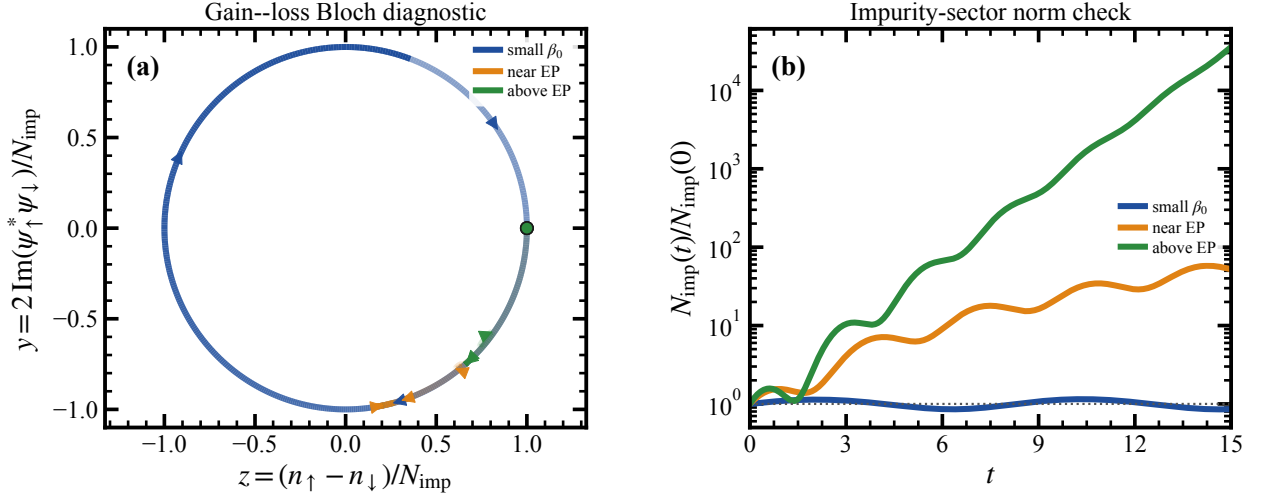


FIG. S1. **Gain-loss Bloch diagnostic for the impurity sector.** Panel (a) shows the normalized (z, y) trajectory for representative values of β_0 below, near, and above the EP. Below the EP the normalized co-decaying trajectory remains close to the unbroken relative-kernel orbit, while close to and beyond the EP the trajectory is progressively distorted. Panel (b) tracks the impurity-sector norm $N_{\text{imp}}(t)/N_{\text{imp}}(0)$ and shows that the above-EP trajectory acquires a strong late-time amplification, whereas the below-EP and near-EP traces remain well controlled over the plotted time window.

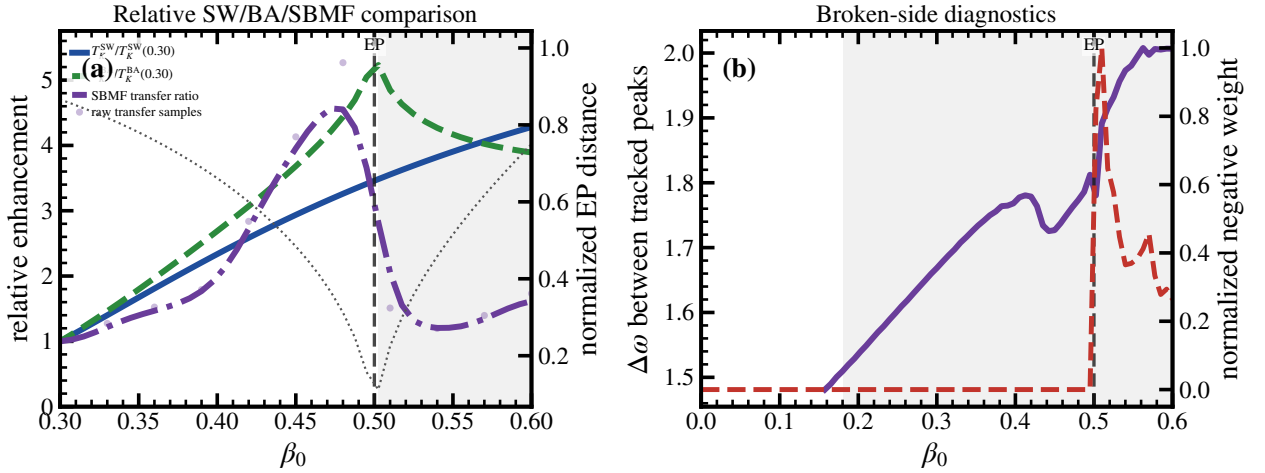


FIG. S2. **Supplementary scale and peak-gap diagnostics near the EP.** Panel (a) compares the analytic SW scale, the frozen BA-kernel scale, and the independent SBMF positive spectral-weight-transfer ratio $\mathcal{W}_{\text{FSR}}^+/\mathcal{W}_{\text{CP}}^+$ as relative enhancement factors normalized to $\beta_{0,\text{ref}} = 0.30$. Because the three quantities have different absolute normalizations, this relative plot is the quantitative comparison used in the revision. The discriminating feature is that the plain SW estimate grows monotonically by construction, whereas the BA-kernel diagnostic and the independent SBMF transfer ratio share a maximum near $\beta_{0,\text{proj}}^{\text{EP}}$; this co-peaking, rather than the overall growth, is the evidence for the EP-assisted redistribution. The SBMF curve terminates at the negative-signed-weight onset at the EP. The dotted grey curve shows the normalized EP-distance diagnostic. Panel (b) shows the peak separation $\Delta\omega$ extracted from the tracked spectral peaks together with the onset of negative signed weight. This panel is used for the broken-side response, where FWHM is no longer a meaningful spectral scale.

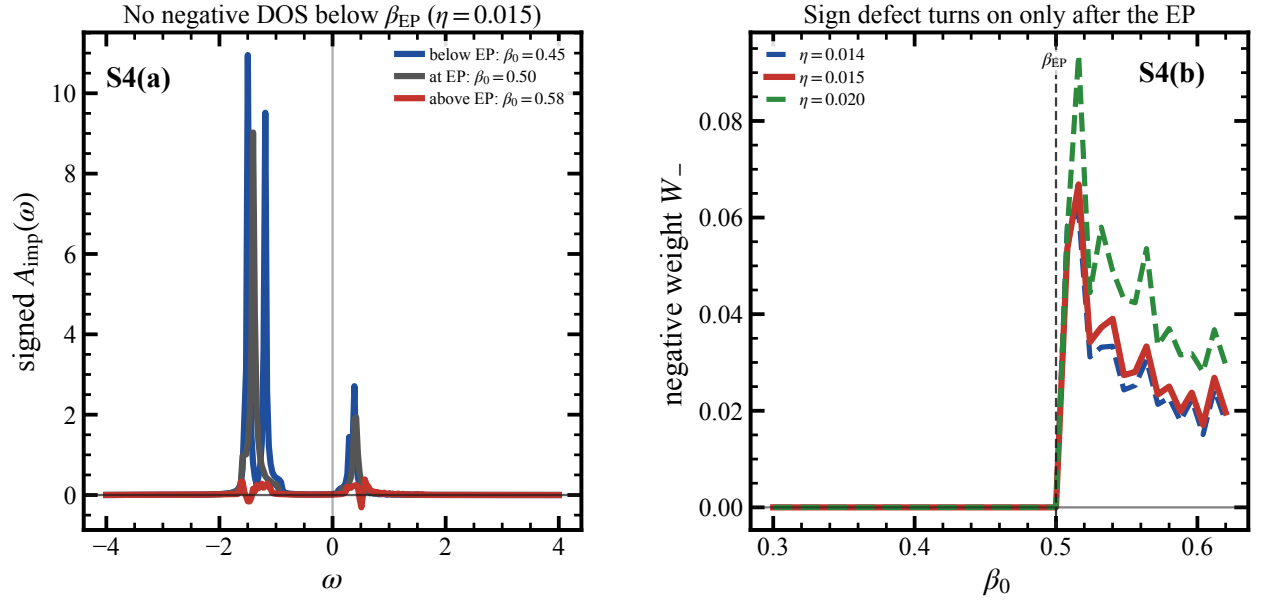


FIG. S3. **Onset of negative signed spectral weight only beyond the EP.** Panel S3(a) compares the signed impurity spectral response below the EP ($\beta_0 = 0.45$), at the EP ($\beta_0 = 0.50$), and above the EP ($\beta_0 = 0.58$) for $\delta_{\text{br}} = 0.015$. No negative signed relative-kernel response appears below the EP, while a sign defect emerges only on the PT-broken side. Panel S3(b) quantifies the integrated negative weight W_- for several broadenings and shows that it remains zero up to the EP and turns on only after $\beta_{0,\text{proj}}^{\text{EP}} \simeq 0.5$. This confirms that the negative signed response is an above-EP effect rather than a broadening artefact.

REPRODUCIBILITY

The numerical scripts, parameter tables, and exported data used for the main and Supplemental figures are included in the accompanying archive and in the version-controlled GitHub repository `VMKPHYSMATH/emergent-pt-dirac-impurity` [5].

-
- [1] A. Kamenev, *Field Theory of Non-Equilibrium Systems*, 2nd ed. (Cambridge University Press, 2023).
 - [2] V. M. Kulkarni and N. S. Vidhyadhiraja, Anderson impurities in edge states with nonlinear and dissipative perturbations, *SciPost Phys.* **19**, 036 (2025).
 - [3] J. M. N. García and L. Wyss, Jordan blocks and the Bethe ansatz I: The eclectic spin chain as a limit, *Nucl. Phys. B* **981**, 115860 (2022).
 - [4] Y. Yi, Y. Qiao, J. Cao, and W.-L. Yang, A novel Bethe ansatz scheme for the one-dimensional Hubbard model, *Nucl. Phys. B* **977**, 115732 (2022).
 - [5] V. M. Kulkarni, Reproducibility code and data for “Emergent PT Symmetry and Exceptional Points in a Driven Dirac Impurity”, GitHub repository (2026).